WASP-36b

R-1830

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_191225` · created 2026-08-03T05:44:57+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458841.3572 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.72 +/- 1.58 [%]
Orbital Inclination (inc)	84.3 +/- 2.88
Airmass coefficient 1 (a1)	11.53 +/- 4.4
Airmass coefficient 2 (a2)	-1.78 +/- 0.58
Scatter in the residuals of the lightcurve fit is	29.5 %
Best Comparison Star	#1 - [210, 180]
Optimal Aperture	12.37
Optimal Annulus	40.69
Transit Duration (day)	0.0296 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.048 vs 0.1368 (deviation 0.59σ)
Duration fit vs published	0.0296 d vs 0.0758 d (deviation 4.82σ)
Mid-time O–C	-2206.2 min
Depth SNR	3.4
Residual scatter	29.5 %

Light curve

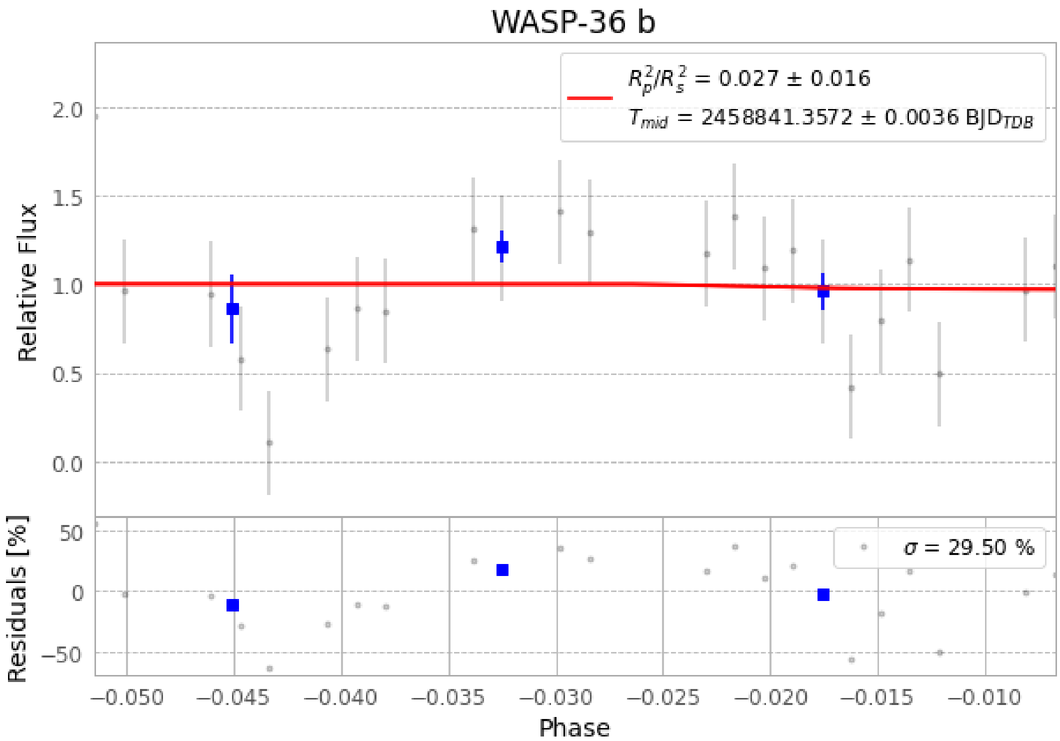

Figure 1 — FinalLightCurve_WASP-36 b_2019-12-25.png (SHA-256 400097c73328cce6…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [253, 437], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f23ae80751f225d5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

36 FITS frames (24 MB), `WASP-361912250727l2.FITS` → `WASP-36191225091212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-191225.log (e166ae449290…), FinalLightCurve_WASP-36 b_2019-12-25.png (400097c73328…), FinalLightCurve_WASP-36 b_2019-12-25.csv (75f3a3a70cf0…)

Compute resources

Wall time 255.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)